

Finite Element Approximation of a Convection-Diffusion-Reaction Equation

Charly Martin Avila

May 18, 2026

Contents

1	Introduction	2
2	Variational Formulation	2
3	Finite Element Discretization	3
3.1	Mass Matrix	3
3.2	Stiffness Matrix	3
3.3	Convection Matrix	4
4	Manufactured Exact Solution	4
5	Implementation	5
6	Numerical Results	5
6.1	Mesh Visualization	5
6.2	Numerical Solution	6
6.3	Exact Solution	7
6.4	Convergence Study	7
7	Connected Components	9
8	Discussion	9
9	Conclusion	9

1 Introduction

The objective of this project is to solve numerically a convection-diffusion-reaction partial differential equation using conforming P1 finite elements on triangular meshes.

The equation considered is

$$-\Delta u + b \cdot \nabla u + cu = f \quad \text{in } \Omega,$$

with homogeneous Dirichlet boundary conditions

$$u = 0 \quad \text{on } \partial\Omega.$$

The implementation is entirely written in Python using:

- NumPy,
- SciPy,
- Matplotlib,
- meshio.

The project reuses the matrix assembly routines implemented during TP3 and extends them to a complete finite element solver.

2 Variational Formulation

We consider the PDE

$$-\Delta u + b \cdot \nabla u + cu = f.$$

Let

$$V = H_0^1(\Omega).$$

We multiply the equation by a test function $v \in V$ and integrate over the domain:

$$\int_{\Omega} (-\Delta u)v + \int_{\Omega} (b \cdot \nabla u)v + c \int_{\Omega} uv = \int_{\Omega} fv.$$

Using integration by parts on the Laplace term:

$$\int_{\Omega} (-\Delta u)v = \int_{\Omega} \nabla u \cdot \nabla v - \int_{\partial\Omega} \frac{\partial u}{\partial n} v.$$

Since $v = 0$ on $\partial\Omega$, the boundary integral vanishes. Therefore, the weak formulation becomes:

Find $u \in H_0^1(\Omega)$ **such that**

$$a(u, v) = \ell(v), \quad \forall v \in H_0^1(\Omega),$$

where

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v + \int_{\Omega} (b \cdot \nabla u) v + c \int_{\Omega} uv,$$

and

$$\ell(v) = \int_{\Omega} fv.$$

3 Finite Element Discretization

We discretize the domain using triangular meshes and P1 Lagrange finite elements. The finite dimensional space is

$$V_h \subset H_0^1(\Omega).$$

A discrete function is written as

$$u_h = \sum_{j=1}^N U_j \phi_j,$$

where:

- ϕ_j are the global P1 basis functions,
- U_j are the nodal coefficients.

The discrete linear system is

$$(K + C + cM)U = F.$$

3.1 Mass Matrix

The global mass matrix is defined by

$$M_{ij} = \int_{\Omega} \phi_i \phi_j.$$

For one triangle T , the elementary matrix is

$$M_T = \frac{|T|}{12} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

3.2 Stiffness Matrix

The stiffness matrix is

$$K_{ij} = \int_{\Omega} \nabla \phi_i \cdot \nabla \phi_j.$$

Since P1 basis functions are affine on each triangle, their gradients are constant on each element.

3.3 Convection Matrix

The convection matrix is

$$C_{ij} = \int_{\Omega} (b \cdot \nabla \phi_j) \phi_i.$$

On one triangle T ,

$$(C_T)_{ij} = \int_T (b \cdot \nabla \phi_j) \phi_i.$$

Since $\nabla \phi_j$ is constant on the triangle,

$$(C_T)_{ij} = (b \cdot \nabla \phi_j) \int_T \phi_i.$$

Using

$$\int_T \phi_i = \frac{|T|}{3},$$

we obtain

$$(C_T)_{ij} = \frac{|T|}{3} (b \cdot \nabla \phi_j).$$

4 Manufactured Exact Solution

The source term is chosen as

$$f(x, y) = e^{(b_x x + b_y y)/2} \sin(p\pi x) \sin(q\pi y).$$

We look for a solution of the form

$$u_{ex} = \alpha f.$$

A direct computation gives

$$(-\Delta + b \cdot \nabla + c)f = \left(\pi^2(p^2 + q^2) + c + \frac{|b|^2}{4} \right) f.$$

Therefore,

$$\alpha = \frac{1}{\pi^2(p^2 + q^2) + c + \frac{|b|^2}{4}}.$$

5 Implementation

The project is structured as follows:

```
src/  
|-- mesh_io.py  
|-- matrices.py  
|-- geometry.py  
|-- load.py  
|-- dirichlet.py  
|-- exact.py  
|-- errors.py  
|-- plotting.py  
'-- connected_components.py
```

The matrix assembly uses sparse matrices from `scipy.sparse`.

Example of assembly loop:

```
for triangle in tri:  
    local_matrix = elementary_matrix(...)  
    assemble into global sparse matrix
```

6 Numerical Results

6.1 Mesh Visualization

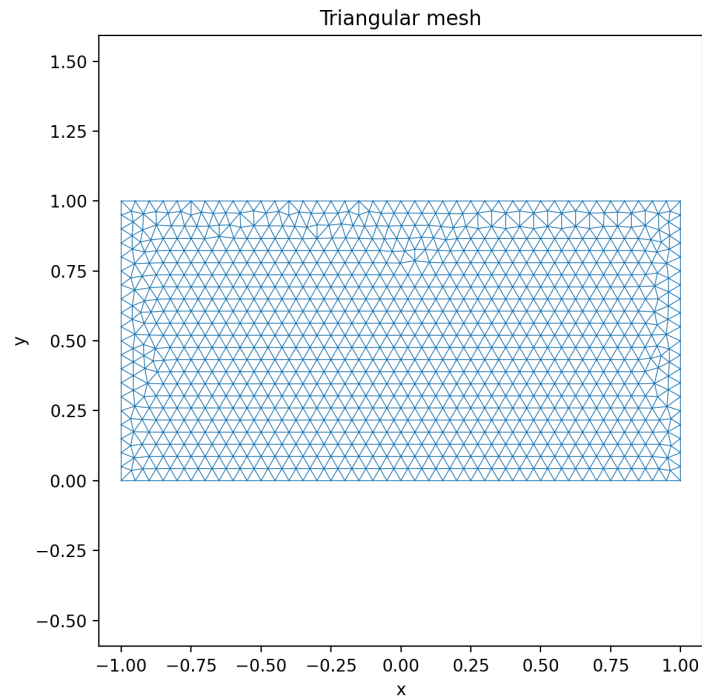


Figure 1: Triangular mesh.

6.2 Numerical Solution

The numerical solution u_h obtained with the finite element method reproduces the main qualitative features of the manufactured exact solution. The solution remains smooth over the computational domain and correctly satisfies the imposed homogeneous Dirichlet boundary conditions on both the exterior boundary and the internal hole.

The deformation of the level sets is influenced by the convection term $b \cdot \nabla u$, which introduces a directional transport effect in the solution.

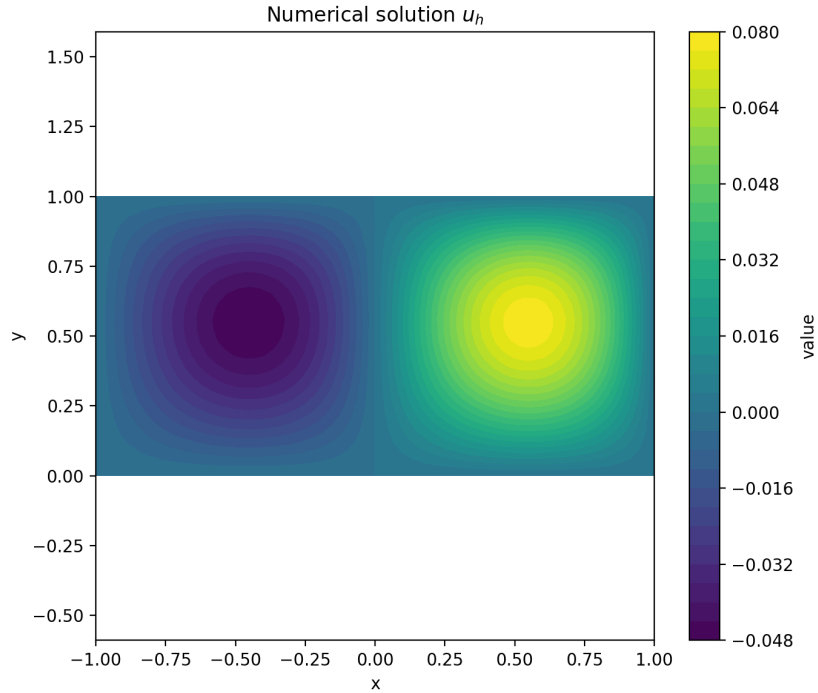


Figure 2: Finite element numerical solution u_h .

6.3 Exact Solution

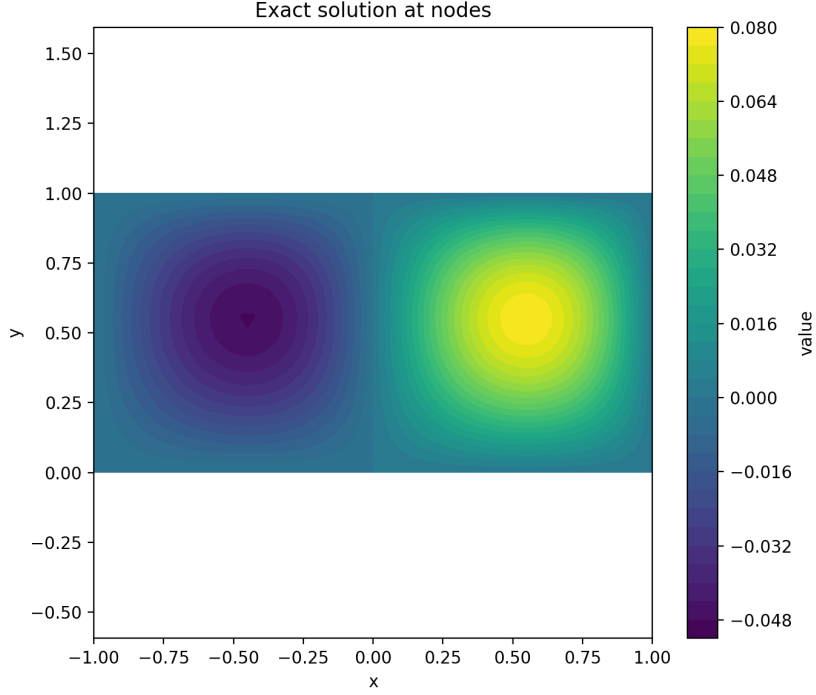


Figure 3: Exact manufactured solution sampled at mesh nodes.

The comparison between the numerical and exact solutions shows a good agreement. The extrema and global structure of the solution are accurately captured by the finite element approximation.

6.4 Convergence Study

The relative errors considered are

$$\frac{\|u_h - \Pi_h u_{ex}\|_{L^2}}{\|\Pi_h u_{ex}\|_{L^2}}$$

and

$$\frac{\|u_h - \Pi_h u_{ex}\|_{H^1}}{\|\Pi_h u_{ex}\|_{H^1}}.$$

The expected convergence rates for P1 finite elements are:

$$L^2 = O(h^2), \quad H^1 = O(h).$$

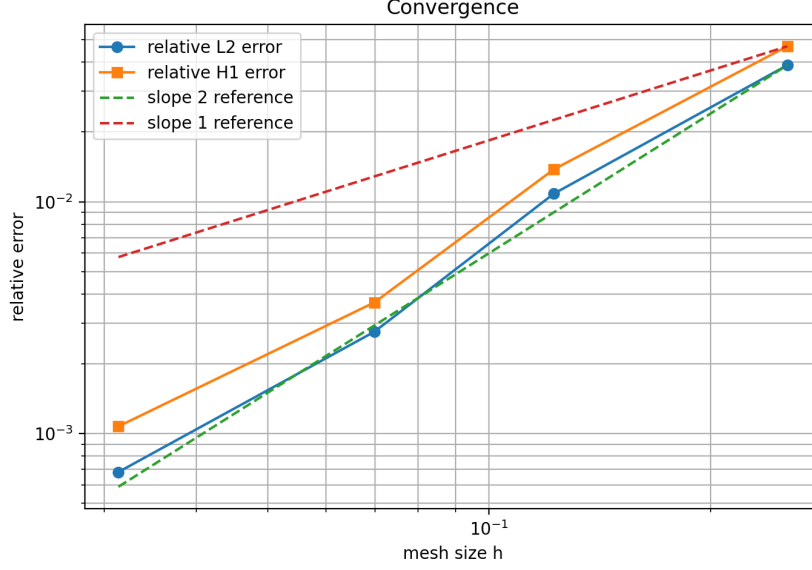


Figure 4: Log-log convergence curves.

The convergence study was performed on a sequence of successively refined meshes of the same domain. The characteristic mesh sizes used in the experiments were approximately

$$h = 0.2, 0.1, 0.05, 0.025.$$

The parameters used for the manufactured solution were

$$p = 1, \quad q = 1, \quad b = (1, 1), \quad c = 1.$$

The graph represents the evolution of the relative L^2 and H^1 errors with respect to the mesh size h on a logarithmic scale.

As expected, the numerical errors decrease when the mesh is refined. This behavior confirms the convergence of the conforming P1 finite element approximation.

The blue curve corresponds to the relative L^2 error, while the orange curve corresponds to the relative H^1 error. The dashed reference lines represent theoretical convergence rates of order

$$O(h^2)$$

and

$$O(h).$$

The results show that the finite element approximation becomes increasingly accurate as the mesh size decreases. The approximately linear behavior of the curves in the log-log plot is characteristic of algebraic convergence rates in finite element methods.

The convergence study therefore validates both the correctness of the matrix assembly procedures and the consistency of the numerical implementation.

Mesh	Mesh size h	Relative L^2 error	Relative H^1 error
square_h0.2	2.543616×10^{-1}	3.882413×10^{-2}	4.682987×10^{-2}
square_h0.1	1.225047×10^{-1}	1.083863×10^{-2}	1.380229×10^{-2}
square_h0.05	6.985550×10^{-2}	2.751216×10^{-3}	3.679815×10^{-3}
square_h0.025	3.135021×10^{-2}	6.827147×10^{-4}	1.074513×10^{-3}

Table 1: Relative errors for successive mesh refinements with $p = q = 1$, $b = (1, 1)$, and $c = 1$.

7 Connected Components

For domains containing holes, the boundary is decomposed into several connected components:

$$\partial\Omega = \Gamma_0 \cup \Gamma_1 \cup \dots \cup \Gamma_J.$$

A graph traversal algorithm (BFS/DFS) is used to identify and label each connected component of the boundary.

This allows us to impose different Dirichlet boundary conditions on each component.

8 Discussion

When the magnitude of the advection vector b increases, the problem becomes convection-dominated.

Numerically, this can lead to:

- directional transport effects,
- boundary layers,
- oscillations on coarse meshes.

This is a classical limitation of the standard Galerkin finite element method.

9 Conclusion

This project extends the finite element matrix assembly routines developed during TP3 to solve a full convection-diffusion-reaction problem.

The numerical results validate:

- the correctness of the assembly routines,
- the expected convergence rates,
- the handling of connected boundary components.

The project also illustrates the influence of convection on the numerical behavior of the finite element method.